

Climate Driven Shifts in Agricultural Land Suitability in California's Central Valley

**A GIS-BASED MULTI-CRITERIA EVALUATION UNDER CURRENT
AND 2050 CLIMATE SCENARIOS**

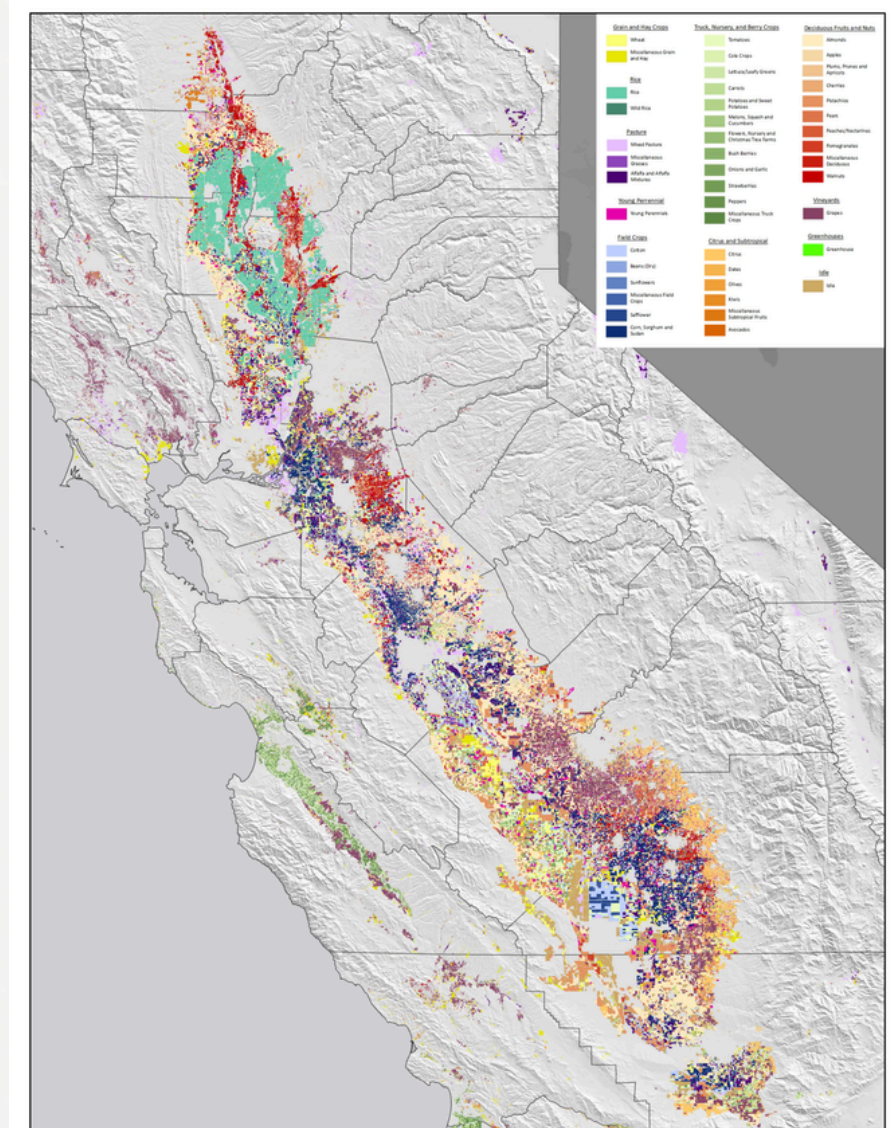
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Background & Motivation

- California produces nearly 50% of U.S. vegetables and over 75% of U.S. fruits & nuts
 - $\frac{3}{4}$ of CA farmland in the Central Valley
- Climate vulnerabilities: droughts, groundwater depletion, salinization, extreme heat
- **Climate suitability mapping** allows for planning, resilience, and water management



Research Problem & Objectives

Problem: Climate change is altering temperature/precipitation
→ affecting crop suitability

Objective: Evaluate spatial shifts in agricultural suitability using GIS-based MCE

Specific goals:

- Map current suitability
- Model future suitability (2050 SSP245 & SSP585)
- Identify areas of gain/loss

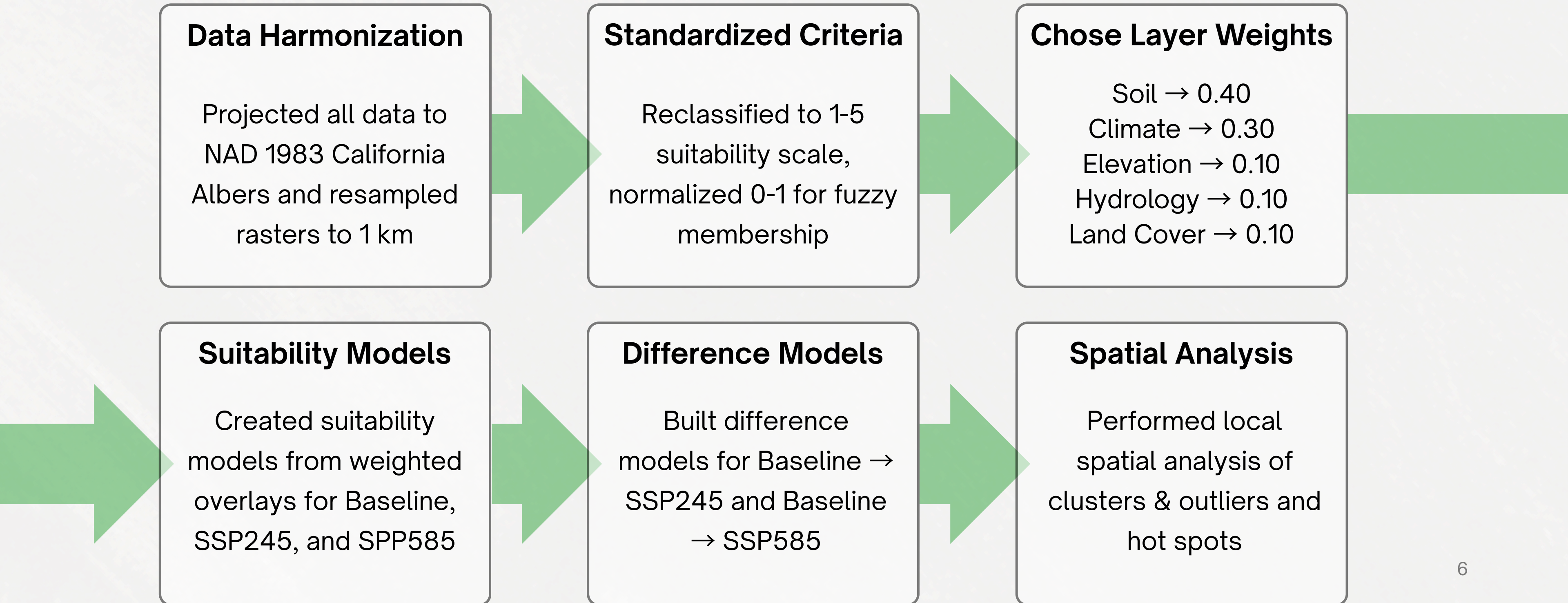
Research Questions

- ① What is the current spatial distribution of suitable agricultural land?
- ② How will climate change alter suitability by 2050 (SSP245 vs. SSP585)?
- ③ Where are suitability gains, losses, or stable zones?
- ④ How can MCE + GIS modeling produce an integrated suitability index?

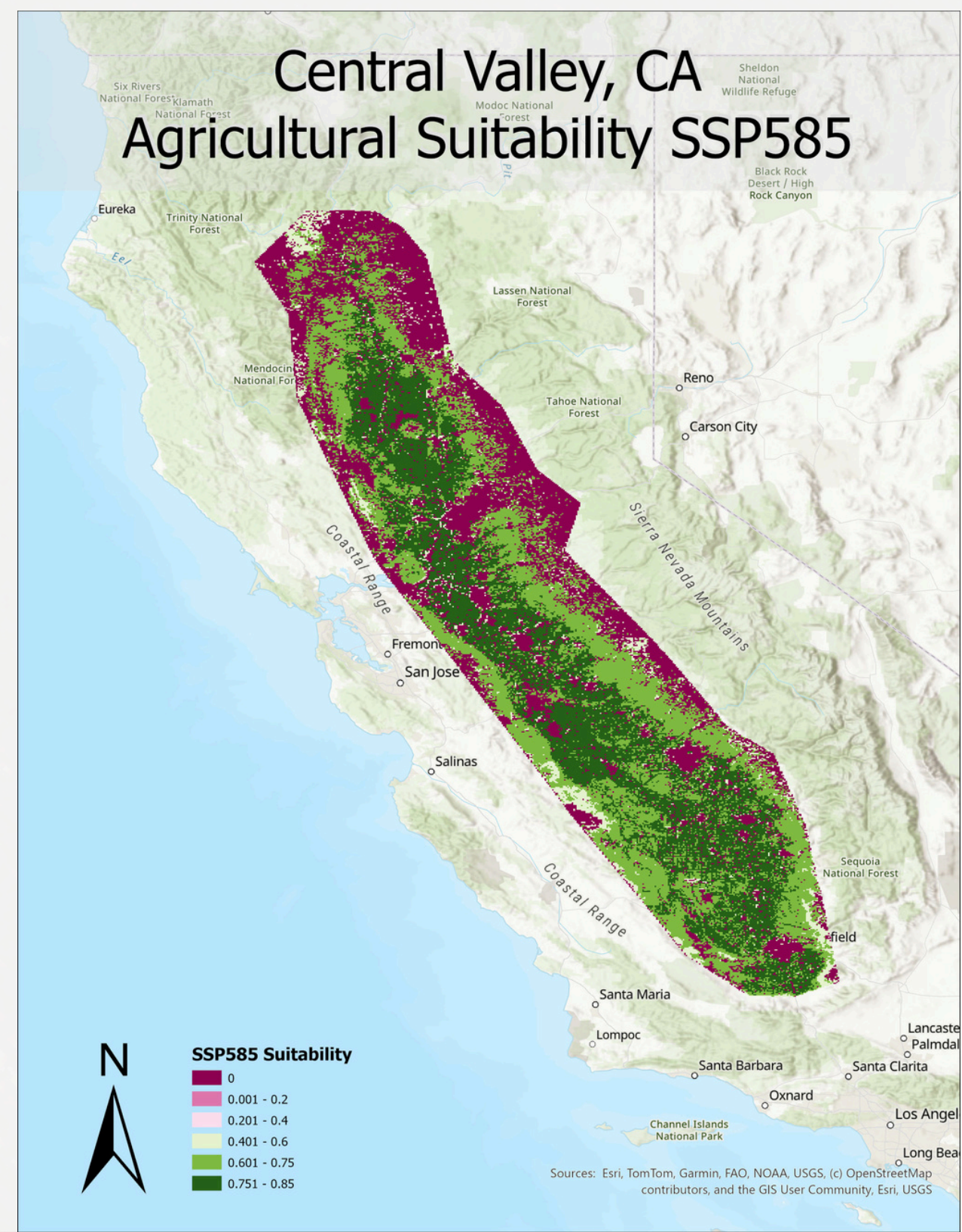
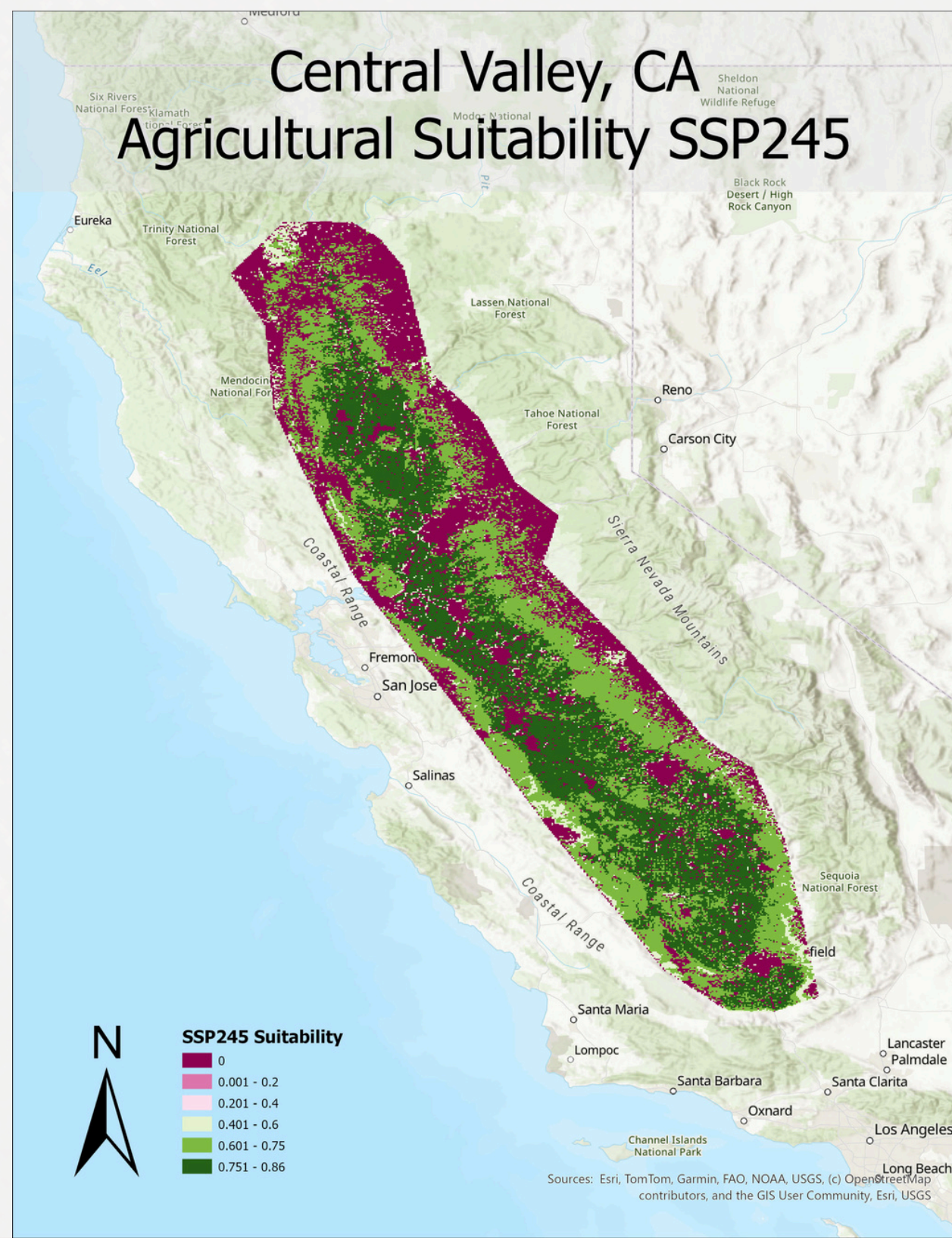
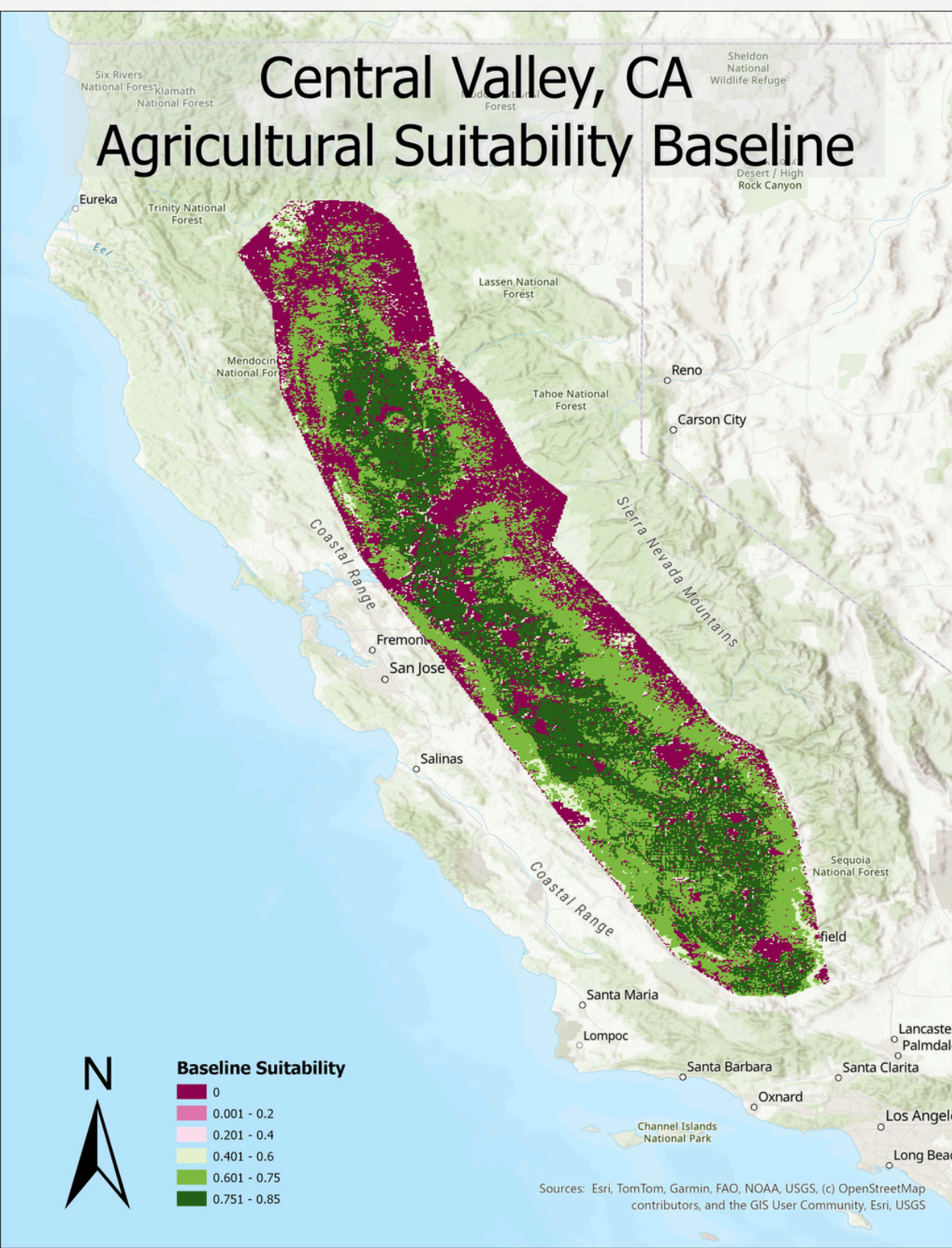
Data Sources

Topic	Key Sources	Features Used
Climate	WorldClim v2.1 (baseline 2000-2020; CMIP6 SSP245/585)	Annual Mean Temp, Max Temp Warmest Month, Annual Precip, Precip Seasonality, Precip Warmest Month
Soils	SoilGrids 2.0	Bulk Density, Clay %, Sand %, Organic Carbon, pH
Land Cover	USGS NLCD 2020	Urban areas, Barren land
Topography	USGS 3DEP DEM	Elevation, Slope, Aspect
Hydrology	NHD 2020	Flowlines, Waterbodies
Boundaries	TIGER/Line	Drew a polygon to represent Central Valley boundary

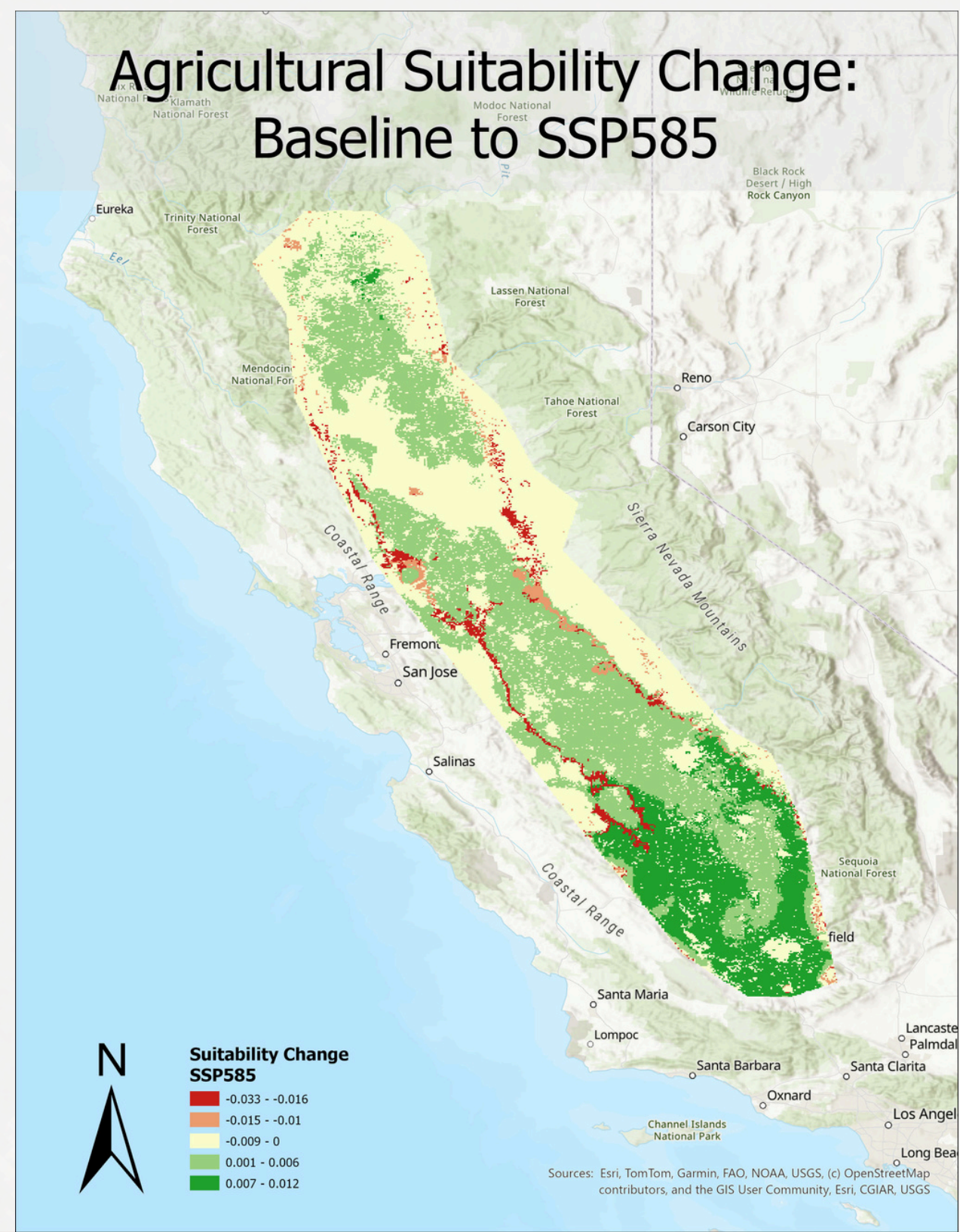
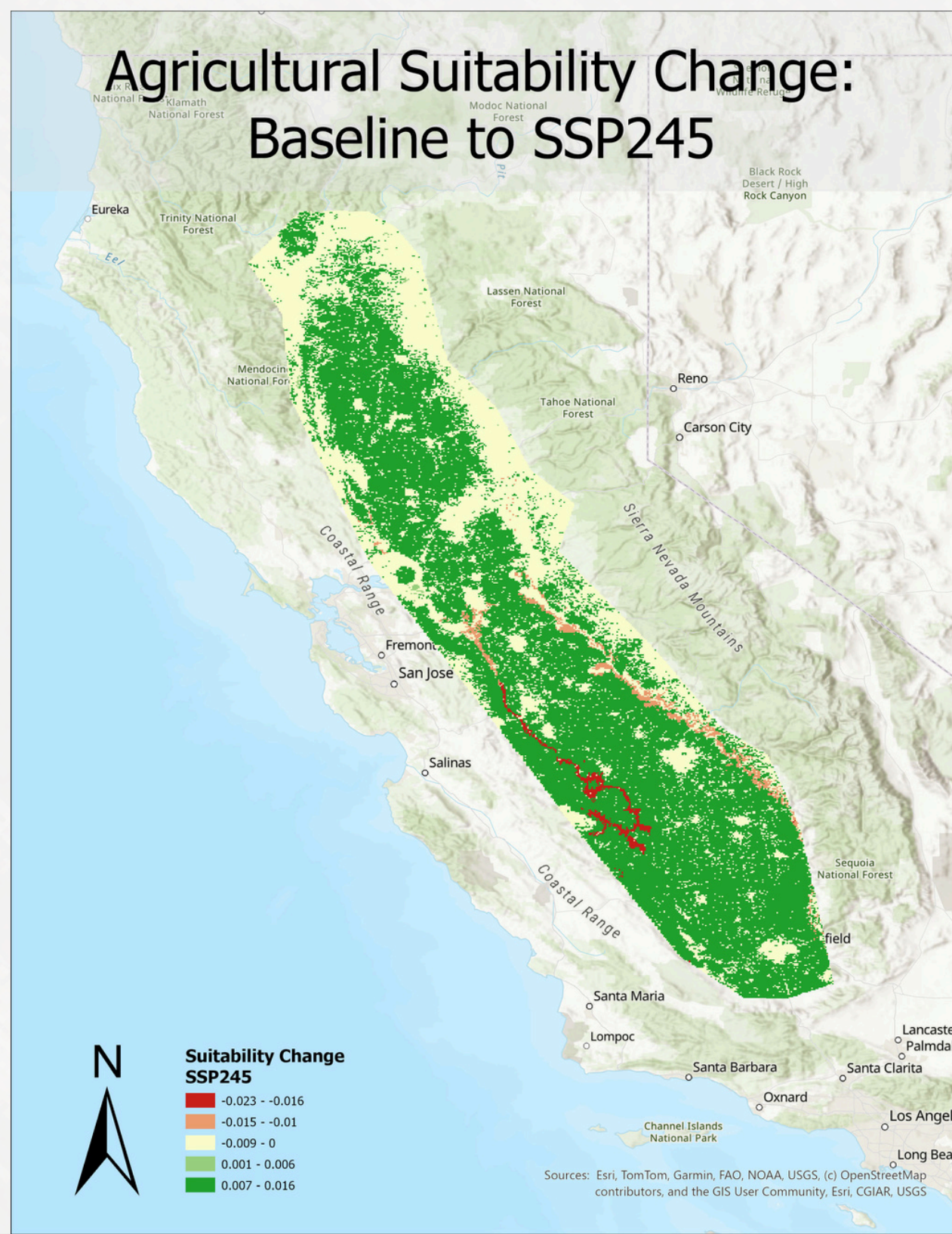
Workflow



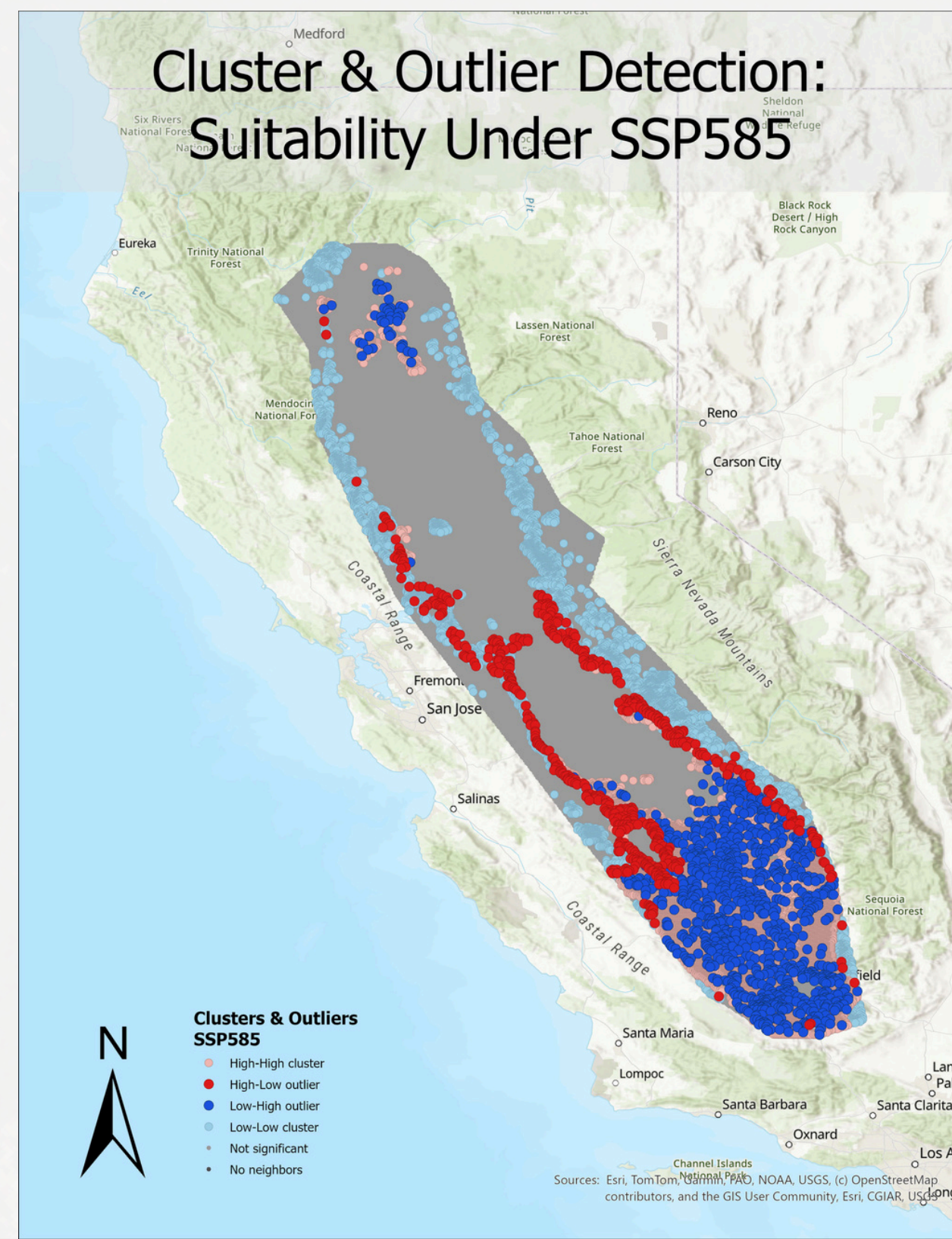
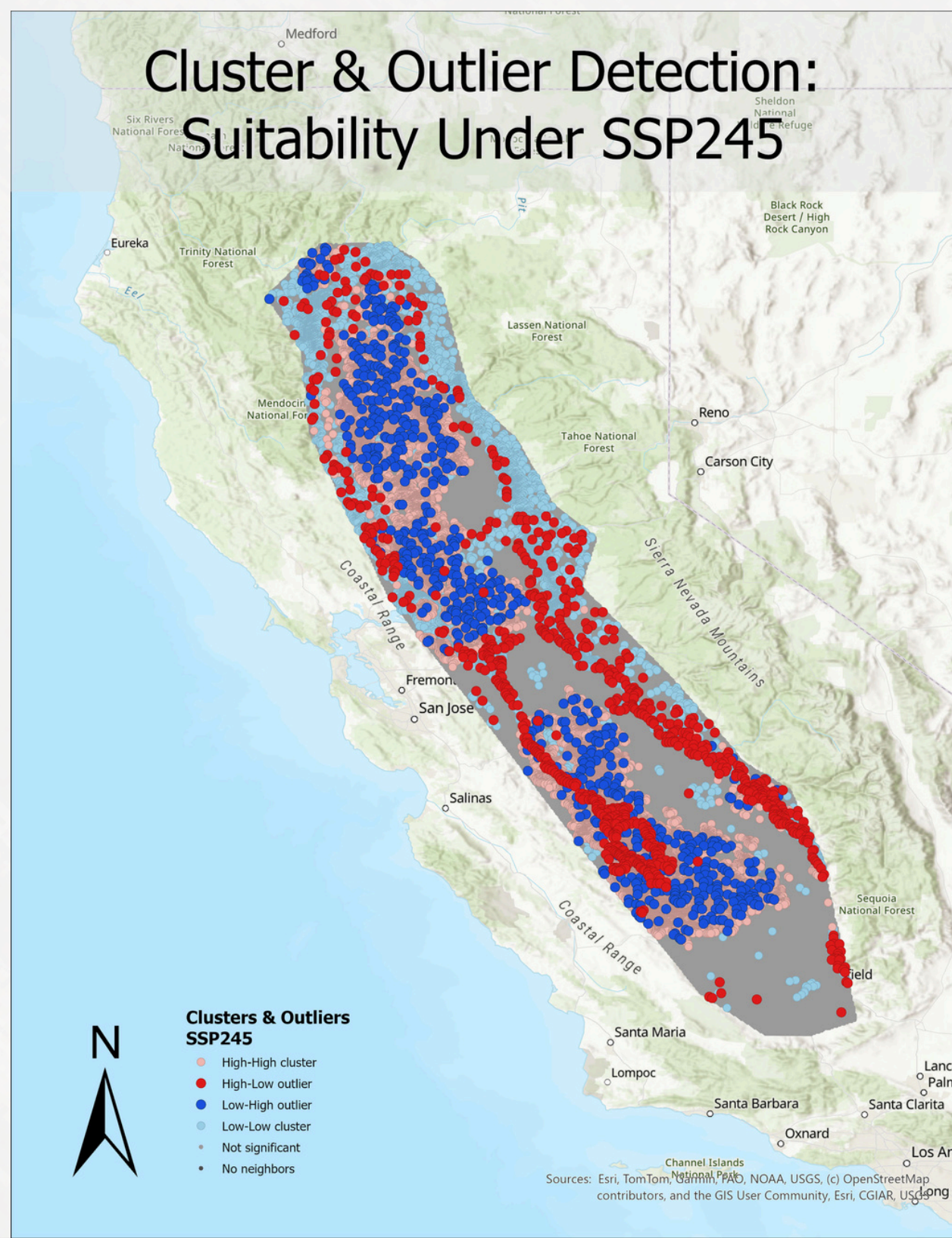
Results & Interpretation: Suitability Models



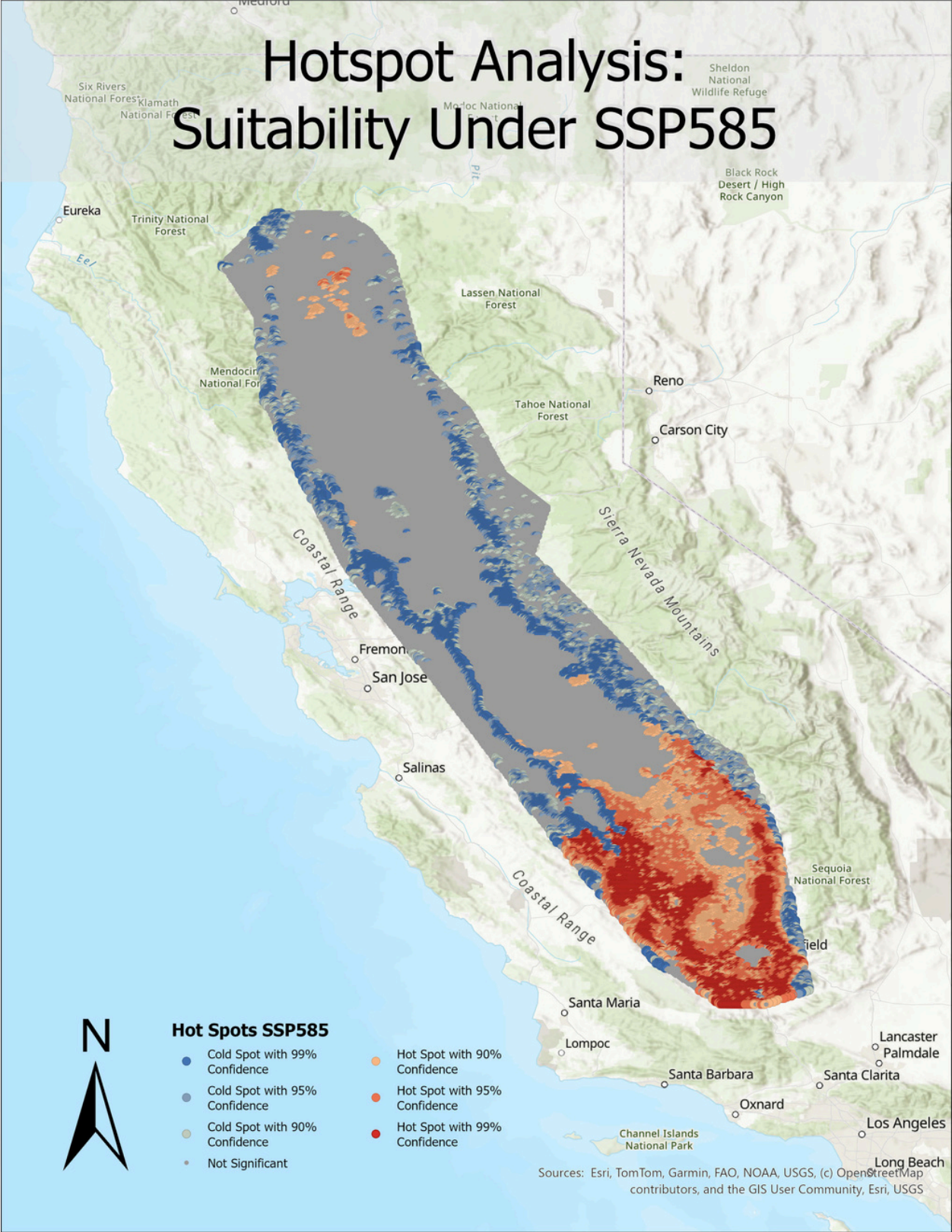
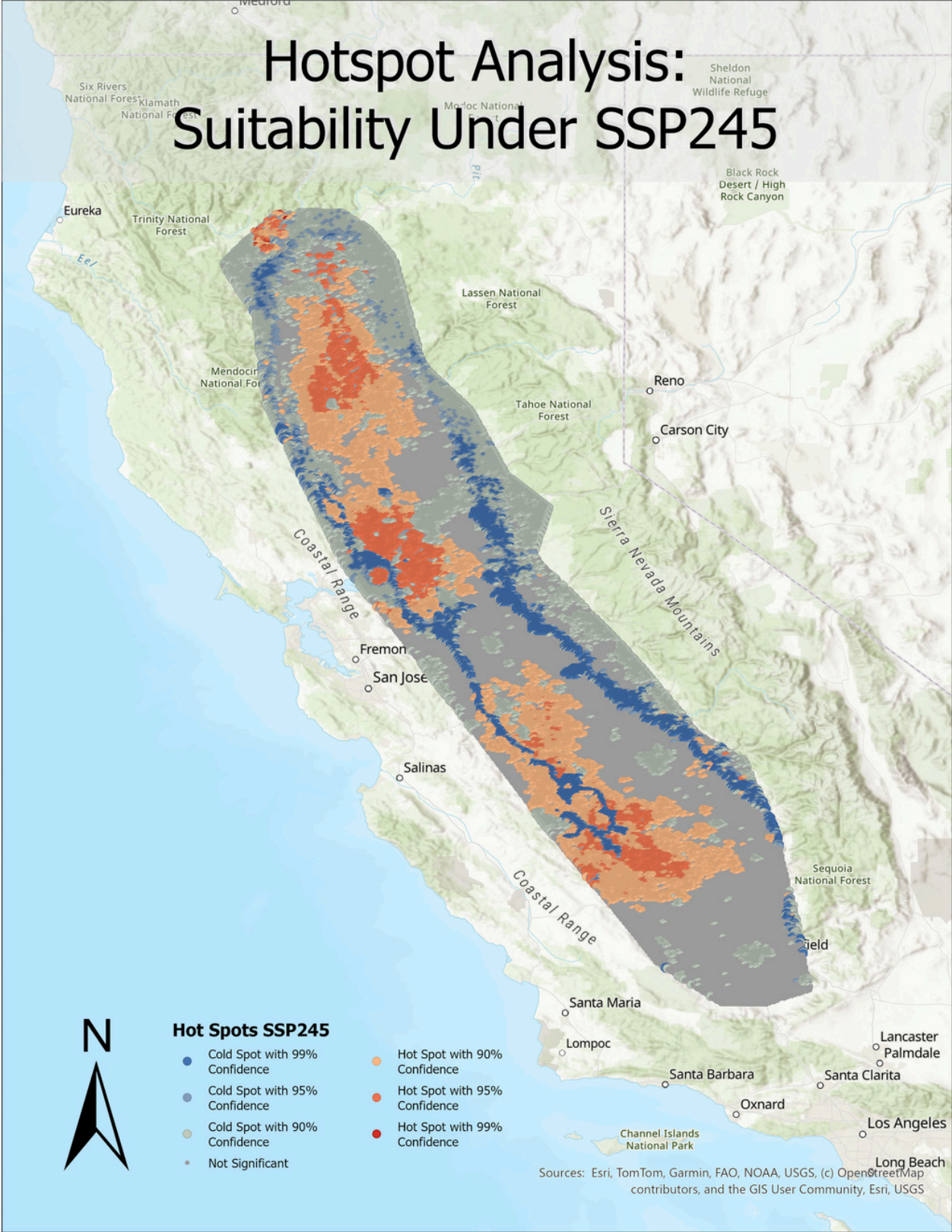
Results & Interpretation: Suitability Change



Results & Interpretation: Clusters & Outliers



Results & Interpretation: Hotspot Analysis



Limitations, Challenges, & Future Work

Challenges:

- Choosing coordinate system to project all data to / work in
- Deciding how to normalize variables for suitability model
- Weighting subjectivity in MCE
- Data resolution differences

Future Work:

- Crop-specific models
- Inclusion of groundwater trends
- Socioeconomic factors (costs, water rights, infrastructure)

References

California Agricultural Production Statistics

Agricultural Land Use in California

A crop map for California's Central Valley.

Central Valley, Britannica